

NEWTON SCHOOL

ALGEBRA 1

ANSWER KEY

Percent Mixture, Percent Change & Graphing Lines

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Date: 2026-05-08

CCSS.MATH.CONTENT.HSA-CED.A.2 / HSA-REI.D.10

Part A · Percent Mixture

Problem 1

6 L of 25% saline + 4 L of 50% saline. New %?

Weighted average

Percent

Salt content: $6 \cdot 0.25 + 4 \cdot 0.50 = 1.5 + 2.0 = 3.5$ L

Total volume: $6 + 4 = 10$ L

Percent: $\frac{3.5}{10} = 0.35 = 35\%$

35%

Watch: The answer is between 25% and 50% — sanity check.

Problem 2

12 lbs of 80% Arabica + 8 lbs of 60% Arabica. New %?

Weighted average

Percent

Arabica: $12 \cdot 0.80 + 8 \cdot 0.60 = 9.6 + 4.8 = 14.4$ lbs

Total: $12 + 8 = 20$ lbs

Percent: $\frac{14.4}{20} = 0.72 = 72\%$

72%

Problem 3

5 mL of 40% acid + 15 mL of 8% acid. New %?

Weighted average

Percent

$$\text{Acid: } 5 \cdot 0.40 + 15 \cdot 0.08 = 2.0 + 1.2 = 3.2 \text{ mL}$$

$$\text{Total: } 5 + 15 = 20 \text{ mL}$$

$$\text{Percent: } \frac{3.2}{20} = 0.16 = 16\%$$

16%

Watch: Larger volume of weaker solution pulls the answer toward 8%, not 40%.

Problem 4

3 L of 45% real fruit juice + 7 L of 15% real fruit juice. New %?

Weighted average

Percent

$$\text{Fruit: } 3 \cdot 0.45 + 7 \cdot 0.15 = 1.35 + 1.05 = 2.4 \text{ L}$$

$$\text{Total: } 3 + 7 = 10 \text{ L}$$

$$\text{Percent: } \frac{2.4}{10} = 0.24 = 24\%$$

24%

Problem 5

20 g of 75% gold + 30 g of 25% gold. New %?

Weighted average

Percent

$$\text{Gold: } 20 \cdot 0.75 + 30 \cdot 0.25 = 15 + 7.5 = 22.5 \text{ g}$$

$$\text{Total: } 20 + 30 = 50 \text{ g}$$

$$\text{Percent: } \frac{22.5}{50} = 0.45 = 45\%$$

45%

Part B · Percent Change

Problem 6 · From 50 to 65

Percent change

Increase

Change: $65 - 50 = 15$

$$\frac{15}{50} = 0.30 = 30\%$$

30% Increase

Problem 7 · From 120 to 90

Percent change

Decrease

Change: $90 - 120 = -30$

$$\frac{|-30|}{120} = \frac{30}{120} = 0.25 = 25\%$$

25% Decrease

Problem 8 · From 25 to 40

Percent change

Increase

Change: $40 - 25 = 15$

$$\frac{15}{25} = 0.60 = 60\%$$

60% Increase

Problem 9 · From 200 to 170

Percent change

Decrease

Change: $170 - 200 = -30$

$$\frac{30}{200} = 0.15 = 15\%$$

15% Decrease

Problem 10 · From 18 to 9

Percent change

Decrease

Change: $9 - 18 = -9$

$$\frac{9}{18} = 0.50 = 50\%$$

50% Decrease

Watch: Always divide by the *original* value (18), not the new value.

Part C · Sketch the Graph

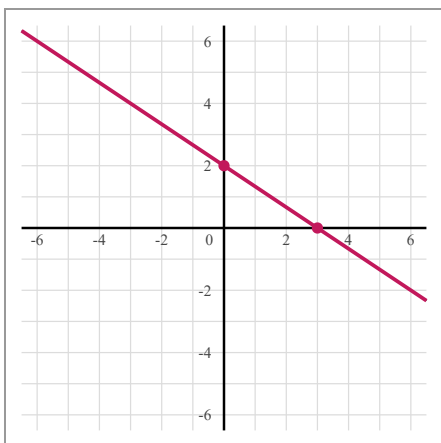
Problem 11 · $2x + 3y = 6$

Intercepts method

x-intercept ($y=0$): $2x = 6 \Rightarrow x = 3 \rightarrow (3, 0)$

y-intercept ($x=0$): $3y = 6 \Rightarrow y = 2 \rightarrow (0, 2)$

Slope: $-\frac{2}{3}$



Points plotted: $(3, 0)$ and $(0, 2)$

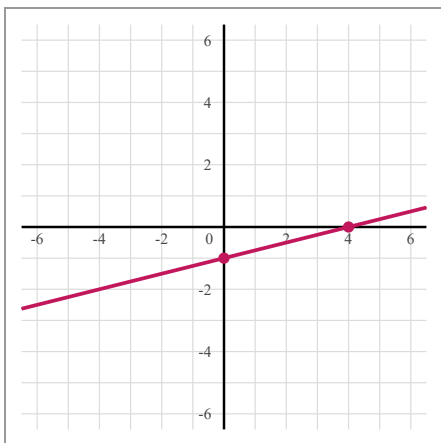
Problem 12 · $x - 4y = 4$

Intercepts method

x-intercept ($y=0$): $x = 4 \rightarrow (4, 0)$

y-intercept ($x=0$): $-4y = 4 \Rightarrow y = -1 \rightarrow (0, -1)$

Slope: $\frac{1}{4}$



Points plotted: $(4, 0)$ and $(0, -1)$

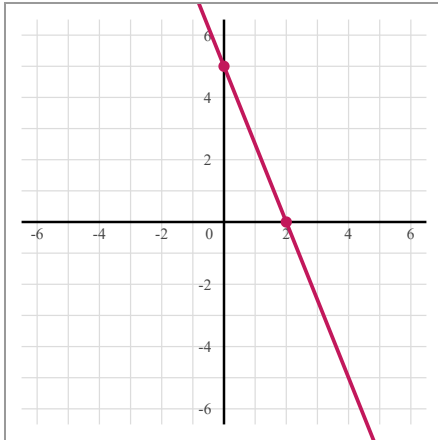
Problem 13 · $5x + 2y = 10$

Intercepts method

x-intercept ($y=0$): $5x = 10 \Rightarrow x = 2 \rightarrow (2, 0)$

y-intercept ($x=0$): $2y = 10 \Rightarrow y = 5 \rightarrow (0, 5)$

Slope: $-\frac{5}{2}$



Points plotted: (2, 0) and (0, 5)

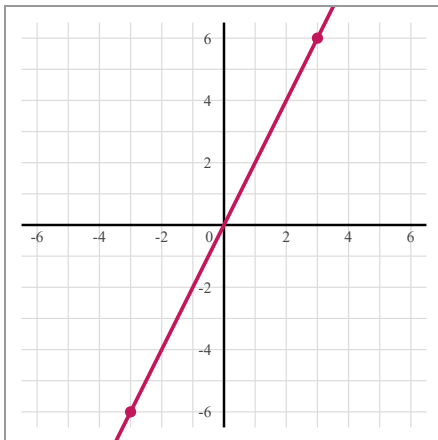
Problem 14 · $6x - 3y = 0$

Through origin

Both intercepts are (0, 0). Need a second point — solve for y: $y = 2x$

Pick $x = 1$: $y = 2 \rightarrow (1, 2)$. Pick $x = -1$: $y = -2 \rightarrow (-1, -2)$

Slope: 2



Line through origin with slope 2: passes through (1, 2), (2, 4), (-1, -2)

Watch: Lines through the origin can't use the intercepts method alone — you must pick another point.

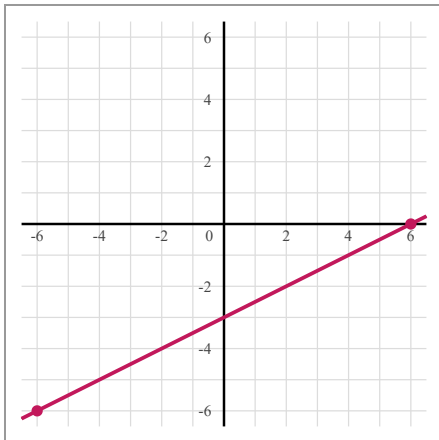
Problem 15 · $y = \frac{1}{2}x - 3$

Slope-intercept form

y-intercept: $-3 \rightarrow (0, -3)$

Slope: $\frac{1}{2}$ — rise 1, run 2

From $(0, -3)$: right 2, up 1 $\rightarrow (2, -2)$. Right 2, up 1 $\rightarrow (4, -1)$. Right 2, up 1 $\rightarrow (6, 0)$.



Points plotted: $(0, -3)$, $(2, -2)$, $(4, -1)$, $(6, 0)$